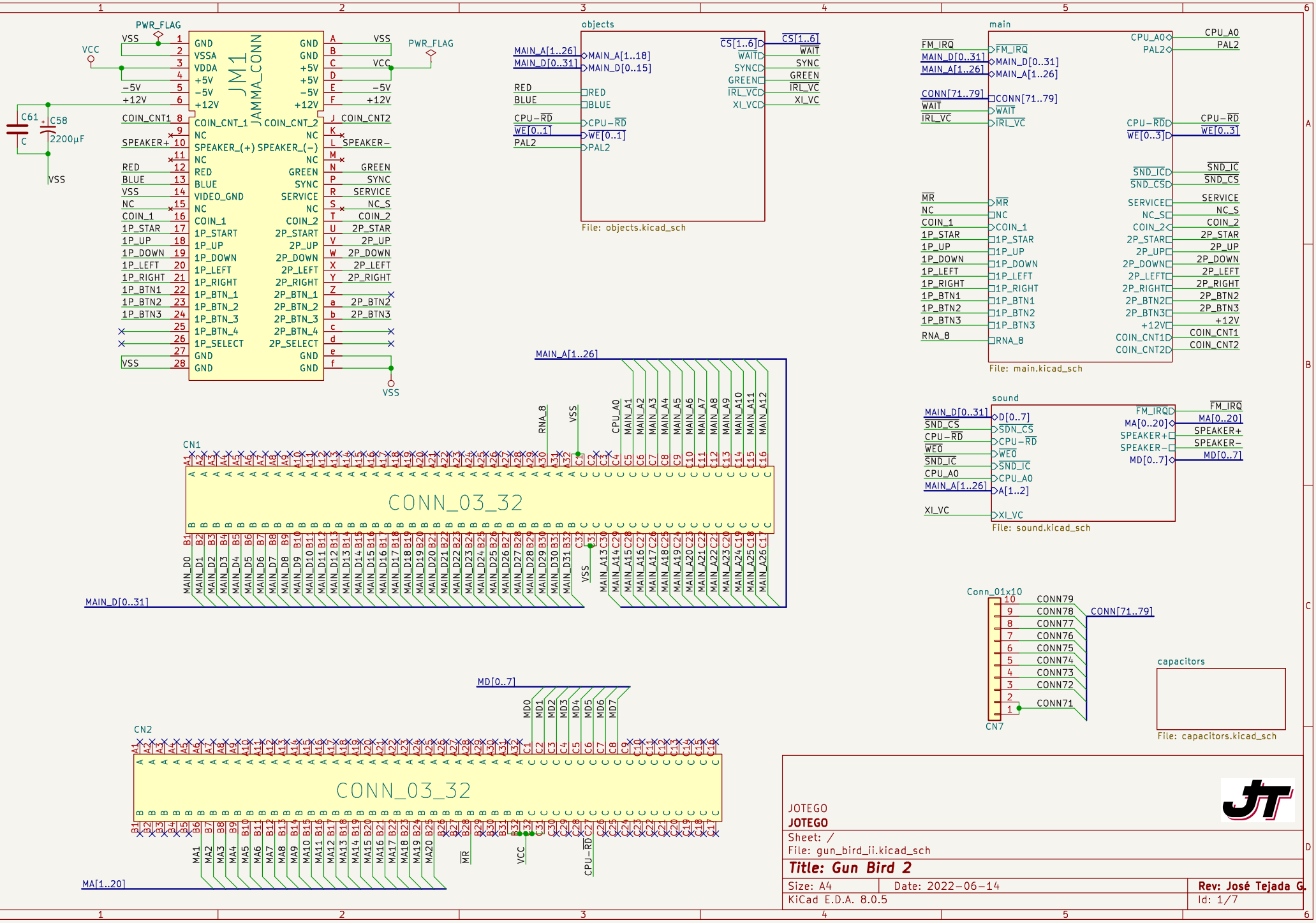

PWR_FLAG
VCC
VSS
C61
C
C58
2200µF
JAMMA_CONN
JM1
GND
VSSA
VDDA
+5V
-5V
+12V
COIN_CNT1
COIN_CNT_1
COIN_CNT_2
COIN_CNT2
NC
SPEAKER+
SPEAKER_(+)
SPEAKER_(-)
SPEAKER-
RED
BLUE
VIDEO_GND
GREEN
SYNC
SERVICE
NC_S
COIN_1
COIN_2
1P_START
2P_START
1P_UP
2P_UP
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
1P_BTN1
2P_BTN1
1P_BTN2
2P_BTN2
1P_BTN3
2P_BTN3
1P_BTN4
2P_BTN4
1P_SELECT
2P_SELECT
objects
MAIN_A[1..26]
MAIN_D[0..31]
MAIN_A[1..18]
MAIN_D[0..15]
CS[1..6]
WAIT
SYNC
GREEN
IRL_VC
XI_VC
CPU-RD
WE[0..1]
PAL2
File: objects.kicad_sch
main
FM_IRQ
CONN[71..79]
CPU_A0
File: main.kicad_sch
sound
MA[0..20]
SPEAKER+
SPEAKER-
MD[0..7]
SND_CS
SDN_CS
WE0
SND_IC
A[1..2]
File: sound.kicad_sch
CN1
CONN_03_32
RNA_8
CN2
MA[1..20]
MR
Conn_01x10
CN7
capacitors
File: capacitors.kicad_sch
JOTEGO
JOTEGO
Sheet: /
File: gun_bird_ii.kicad_sch
Title: Gun Bird 2
Size: A4
Date: 2022-06-14
KiCad E.D.A. 8.0.5
Rev: José Tejada G.
Id: 1/7

U22
PS6406B_98080AFB
VIDEO CONTROLLER

RAMD[0..15]
RAM[0..13]
MAIN_A[1..18]
D[0..31]
MAIN_D[0..15]
VC[0..21]
CS[1..6]
BLUE[0..7]
GREEN[0..7]
RED[0..7]
WE[0..1]
XI(Connection_clock_33.8688MHz)
57.2727MHz
IQXO-70

U20 GLT725608-12J3 TSOP RAM 32K x 8BIT
U21 GLT725608-12J3 TSOP RAM 32K x 8BIT
U30 74LS244
U31 74LS244
U32 74LS244
RN7 LN1028
RN8 LN1028
RN9 LN1028
U38D 74LS14
U38B 74LS14
Q1 Q_NPN_C1815
Q2 Q_NPN_C1815
Q3 Q_NPN_C1815
Q4 Q_NPN_C1815
R2 223Ω
R4 102Ω
R5 332Ω
R6 332Ω
R7 332
R11 471Ω
R12 471Ω
R13 471Ω
R14 471Ω
R17 330Ω
R18 330Ω
R19 330Ω
R20 330Ω
MSB 1 kOhm
LSB 128 kOhm
XU23 E/B OUT GND V+

74LS08 U37E
74LS14 U38G

memories_vc
File: memories_vc.kicad_sch

JOTEGO
JOTEGO
Sheet: /objects/
File: objects.kicad_sch
Title: Gun Bird 2
Size: A3 | Date: 2022-06-14 | **Rev: José Tejada G.**
KiCad E.D.A. 8.0.5 | Id: 2/7

VC[0..21]
CS[1..6]
OE_ROM
D[0..31]

U3 U4 U5 U6 U7 U8 U10 U11 U12 U13 U14 U15

LHMN5KUG-9835-38
ROM 8Mx8-4Mx16

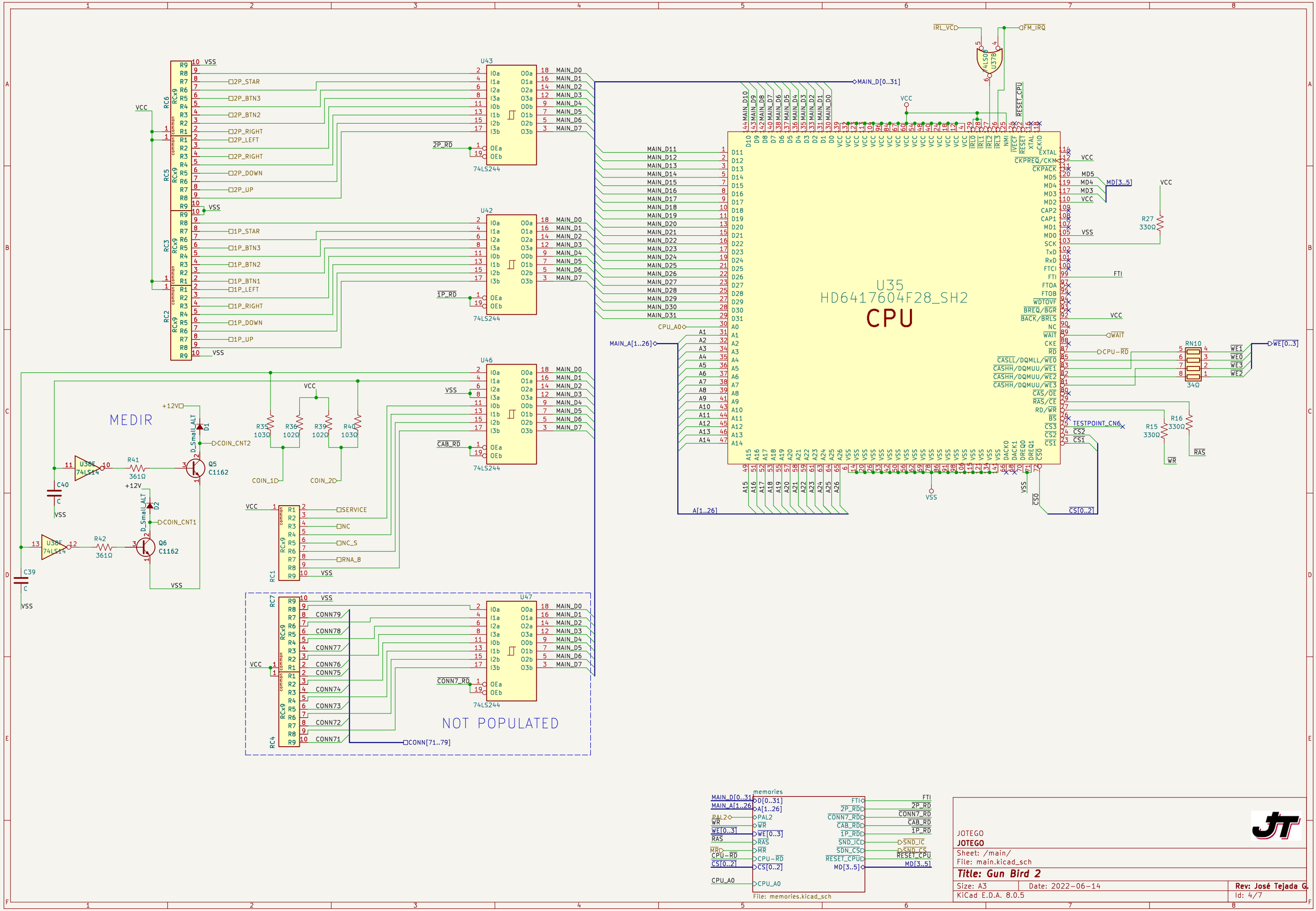
NOT POPULATED

JOTEGO
JOTEGO

Sheet: /objects/memories_vc/
File: memories_vc.kicad_sch

Title: Gun Bird 2

Size: A3	Date: 2022-06-14	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 3/7


U35
HD6417604F28_SH2
CPU
U43
U42
U46
U47
74LS244
MAIN_D[0..31]
MAIN_A[1..26]
MD[3..5]
WE[0..3]
CS[0..2]
FTI
WAIT
RESET_CPU
CPU_A0
CPU-RD
2P_RD
1P_RD
CAB_RD
CONN7_RD
2P_STAR
2P_BTN3
2P_BTN2
2P_RIGHT
2P_LEFT
2P_DOWN
2P_UP
1P_STAR
1P_BTN3
1P_BTN2
1P_BTN1
1P_LEFT
1P_RIGHT
1P_DOWN
1P_UP
SERVICE
NC
NC_S
RNA_8
COIN_CNT1
COIN_CNT2
COIN_1D
COIN_2D
CONN[71..79]
MEDIR
NOT POPULATED
TESTPOINT_CN6
IRL_VCC
FM_IRQ
memories
File: memories.kicad_sch
JOTEGO
JOTEGO
Sheet: /main/
File: main.kicad_sch
Title: Gun Bird 2
Size: A3
Date: 2022-06-14
KiCad E.D.A. 8.0.5
Rev: José Tejada G.
Id: 4/7

PROG L
PROG H
NOT POPULATED
M27C4002
ROM 4Mb
MB3771
93C56X
93LC56
EEPROM
Microwire Serial
74LS273
flip flops
74LS08
74LS14
74LS32
74LS244
74LS138
PALACE16V8H-25
LHMN5KUG-9835-38
ROM 8Mx8-4Mx16
OKI-M514260C-60J
RAM 16BIT
Jumper_4
Jumper_2_Open
address decoder
D[0..31]
A[1..26]
MD[3..5]
CS[0..2]
WE[0..3]
SND_IC
RESET_CPU
CPU-RD
AD_O4
CLK_AD
SDN_CS
PAL2
FTI
1P_RD
2P_RD
CONN7_RD
CAB_RD
CPU_A0
JOTEGO
JOTEGO
Sheet: /main/memories/
File: memories.kicad_sch
Title: Gun Bird 2
Size: A3
Date: 2022-06-14
Rev: José Tejada G.
KiCad E.D.A. 8.0.5
Id: 5/7

MA[0..20]

VCC

A[1..2]

MD[0..7]

U9

LHMN5KUG-9835-38

ROM 8Mx8-4Mx16

MA1 MA2 MA3 MA4 MA5 MA6 MA7 MA8 MA9 MA10 MA11 MA12 MA13 MA14 MA15 MA16 MA17 MA18 MA19 MA20 MCS0

VSS MRD

MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7

MA0

U19

YMF278B-F

SINTHETIZER FM WAVE

CPU_A0 SDN_CS CPU-RD WEO SND_IC

Connection clock 33.8688MHz

NOT CONNECTED NORMALLY

D[0..7]

MA0 MA1 MA2 MA3 MA4 MA5 MA6 MA7 MA8 MA9 MA10 MA11 MA12 MA13 MA14 MA15 MA16 MA17 MA18 MA19 MA20

FM_IRQ CLKOUT BCO LRO WCO MRD MCS0

XI_VCO

74LS08 U37D

R26 102Ω

U45

YAC513-M

2 CHANNEL FLOATING CONVERTER

VSS BCO WCO LRO DO2 VSS SWIN MP

SEL CLK WDCK LRCK DIN TST1 TST2 SWIN MP

CH1 CH2 CV AOUT

4741-JCR U44C

4741-JCR U44D

C43

NOT POPULATED

R31 361Ω

R33 361Ω

C46 10µF

C41

R32 361Ω

INPUT_A

4741-JCR U44A

R34 361Ω

C42

4741-JCR U44B

R30 221Ω

R38 221Ω

C44 4.7µF

C45 4.7µF

U48

HA13118

18 W BTL Audio Power Amplifier

Amp1 Amp2

ASO_Protect Surge_Protect TSD

RV1 R_POT_US

C72 4.7µF

C71 100µF

C70 100µF

C62 100µF

C64 100µF

C60

C59

C63 100µF

C69 47µF

R37 221Ω

R29 221Ω

N.U N.U

SPEAKER+ SPEAKER−

NOT POPULATED

C52 µF

INPUT_A

C80 µF

R23 Ω

C48 µF

PWR_FLAG

R21 Ω

R28 Ω

U39

YAC516-M

TST1 PDIN IC MCLK CKS BICK SDATA LRCK VREFH VREFL NC NC

AVDD(+5V) AVSS

AOUTR AOUTL VCOM TST2 NC NC NC NC NC NC NC NC

C49 C50 C51

C53 µF

U44E

4741-JCR

V+ V−

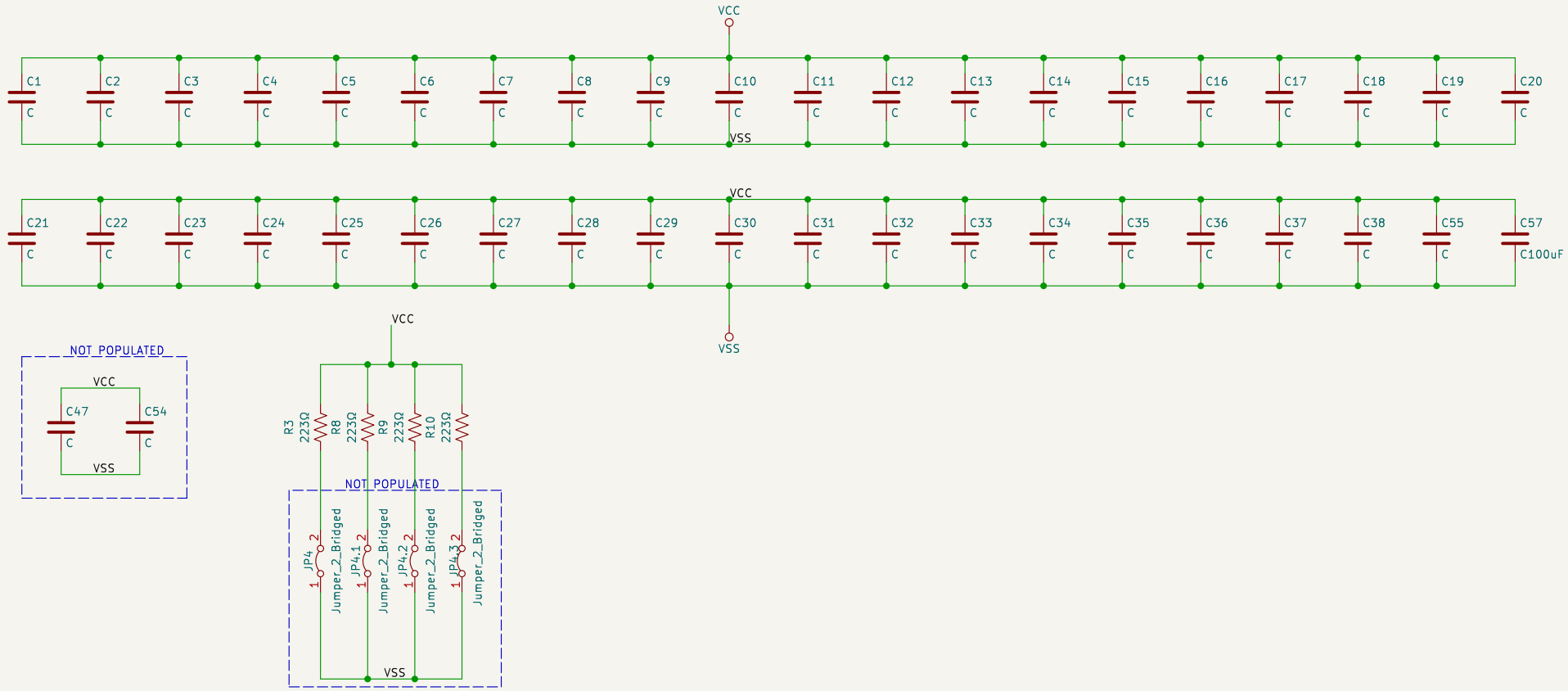
JOTEGO
JOTEGO

Sheet: /sound/
File: sound.kicad_sch

Title: Gun Bird 2

Size: A3 | Date: 2022-06-14 | **Rev: José Tejada G.**

KiCad E.D.A. 8.0.5 | Id: 6/7



JOTEGO
JOTEGO

Sheet: /capacitors/
File: capacitors.kicad_sch

Title: Gun Bird 2

Size: A4	Date: 2022-06-14	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 7/7